

The Refrigeration Cycle

Four components in a loop. Name what each one does to the refrigerant and every component question answers itself.

Air conditioning does not make cold. It moves heat from inside to outside, using a fluid that boils at a convenient temperature. The whole system is one loop, and each component does one thing to that fluid.

Refrigerant	The substance that produces a cooling effect by absorbing heat as it vaporizes. Boiling absorbs heat — that is the entire principle.
Compressor	The component that creates a flow of refrigerant from one part of the system to the other. It is the pump of the loop.
Condenser	Liquefies the refrigerant gas by cooling it. Condenser condenses, and it releases heat outdoors.
Evaporator	Takes heat from the air around it and brings it to the refrigerant. Evaporator evaporates, and it absorbs heat indoors.

YOU HAVE MISSED THIS 2 TIMES

A _____ is an air conditioning component that creates a flow of refrigerant from one part of the system to the other.

Compressor.

YOU HAVE MISSED THIS

_____ is a substance that produces a cooling effect by absorbing heat as it vaporizes.

Refrigerant.

The two names are the answers. If the component cools a gas into a liquid it is condensing; if it lets a liquid boil into a gas it is evaporating.

Refrigerant is the category, Freon is a brand

Freon is the substance used in the past to produce cooling by absorbing heat. When a question wants the general term, it wants refrigerant.

How the equipment is packaged and sized

The numbers here

One ton of air conditioning

12,000 BTU per hour

Packaged unit	What an air conditioner and gas furnace combined on a roof is called. Everything in one cabinet, outdoors.
Split system	The alternative: condenser outside, evaporator inside, joined by refrigerant lines.

Evaporative cooler

Cools by evaporating water into the air. Most likely found in the Southwest, because it only works where the air is dry enough to take up moisture.

YOU HAVE MISSED THIS

When an air conditioner and gas furnace are combined on a roof, this is called:

A packaged unit.

Say it back

Cover the page. Answer each one out loud.

1. What does an air conditioner actually do, if it does not make cold?
2. Name the four components and what each does to the refrigerant.
3. Which absorbs heat and which releases it?
4. Refrigerant or Freon — which is the category and which is the brand?
5. How many BTU per hour is one ton?
6. What is a packaged unit, and where would you find an evaporative cooler?